

Predicting ^1H and ^{13}C Nuclear Magnetic Resonance Chemical Shifts via Hartree-Fock Self Consistent Field Methods

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Abstract—Nuclear Magnetic Resonance (NMR) spectroscopy is a ubiquitous technique in organic chemistry used to both evaluate the structure of individual molecules and assess the composition of mixtures of molecules. In this report, we aim to produce accurate predictions of chemical shifts using computationally cheap methods that are suitable for execution on a modern laptop in near real-time. To this end, we will draw upon various Hartree-Fock (HF) based techniques first published in the 1960s and 1970s to develop accurate methodologies for predicting both ^1H and ^{13}C NMR spectra. We will first seek to reproduce the results of Takaishi et. al's 1974 model for ^{13}C spectra of linear and branching hydrocarbons achieving a ρ of about 0.93. Additionally, we apply the same methodology to an implementation of the INDO model, and compare its accuracy up to the linear regression fit. Lastly, we apply Popel et. al's 1962 model to predict ^1H NMR spectra of the same class of molecules. We attain a ρ of 0.834 with this method.

1. INTRODUCTION

Nuclear Magnetic Resonance (NMR) Spectroscopy is a ubiquitous technique in organic chemistry used to both evaluate the structure of individual molecules and assess the composition of mixtures of molecules. In NMR a strong primary magnetic field is applied to a sample along with a weak, secondary, oscillating magnetic field. The secondary field's oscillations induce a resonance in odd-nucleon nuclei, i.e. those that possess an intrinsic spin and hence magnetic dipole moment.

For a given nucleus, the peak resonance frequency is determined not just by configurations of the field and the nucleus, but the shape of the cloud of electron dipoles that encompass and shield it. Therefore, for the same species of nuclei exposed to identical magnetic fields, the NMR spectra are entirely differentiated by the shape of the electron cloud. This in turn corresponds to the local molecular structure of the nuclei's parent molecule. Consequently, NMR has become an important tool for assessing the structural components of a molecule.

2. OBJECTIVES

In this report, we aim to produce accurate predictions of chemical shifts from NMR spectroscopy using computationally cheap methods that are suitable for execution on a modern laptop in near real-time. To this end, we will draw upon various Hartree-Fock (HF) based techniques first published in the 1960s and 1970s to develop accurate methodologies for predicting both ^1H and ^{13}C NMR spectra.

2.1. Reproduction of Takaishi 1974

We will first seek to reproduce the results of Takaishi et. al's [5] 1974 model for predicting ^{13}C chemical shifts in paraffins. Takaishi's approach uses a highly simplified, non-perturbative CNDO/2 method to individually compute the mean diamagnetic (σ_d) and paramagnetic (σ_p) components of the shielding tensor. As such, it is limited in scope to specifically calculating chemical shifts of paraffins.

Specifically, we will try to reproduce the chemical shifts listed in table (1) of Takashi's paper using our own implementation of CNDO/2, to be assessed via Pearson's ρ and MAE. Additionally, we will compare our results with Takaishi's reported experimental values. Lastly, we will apply the same

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methodology to an implementation of the INDO model, and compare its accuracy up to a linear regression fit.

2.2. Extension of Popel 1962 with Perturbative Paramagnetic Term

Second, we aim to extend Popel’s 1962 general method for calculating the magnetic shielding tensor of an atom [1]. In this paper, Popel applies multiple approximations to a GIAO perturbative theory to arrive at a real-valued, non-perturbative expression for σ . As with Takaishi, this expression can also be decomposed in diamagnetic and paramagnetic terms. We will keep the diamagnetic term, but replace the paramagnetic terms with our own perturbative approach to arrive at an approximation for the ^1H NMR spectra of various molecules, using both CNDO/2 and INDO in our calculations. These results will be compared with experimental values in terms of both Pearson’s ρ and MAE.

3. METHODS

3.1. Direct Inversion in the Iterative Subspace

Direct Inversion in the Iterative Subspace (DIIS) is a numerical method of convergence that consistently outperforms fixed-point iteration in terms of both rate of convergence and sensitivity to initial conditions. Unlike the fixed-point method, which considers only the previous iteration when calculating the next, DIIS is informed by the previous n iterations.

More explicitly, DIIS assigns an error vector (ϵ_i) to each of the past n iterations, and seeks to find a linear combination of the previous errors such that the next iteration’s error is minimized. This is constrained by the condition that the coefficients of the linear combination must sum to 1 (or any other arbitrary non-zero number). Assuming perfect local linearity, the error vector that will minimize the results is given by the following equation:

Without DIIS, the SCF cycle will often fail to converge for molecules of even modest size. For example, we find that our SCF cycle will not converge for n-hexane if we use the fixed-point method. Further discussion can be found in section 4.

3.2. Intermediate Neglect of Differential Overlaps

The second step we took improve the quality of our results was to implement INDO as an alternative to our existing CNDO/2 model. INDO makes the same simplifying assumption of orthogonal atomic orbitals (AOs) as CNDO. Consequently, like CNDO, we can set the overlap matrix $S = I$ for the purposes of the SCF cycle[2], resulting in the simplified Roothaan-Hall equation of

$$FC = CE \tag{2}$$

Therefore, like CNDO, INDO must introduce the approximation of spherically symmetric orbitals for the purpose of computing the two-center electron integrals, constraining them to be identical for any two orbitals from the same atom. However, unlike CNDO, INDO notes that AOs of the same atom are already orthogonal to each other. As such, their block within the overlap matrix S is already equivalent to I within that subspace, and we can recover unique values for one-center (i.e. single atom) electron integrals without losing the simplifications attained in equation 2.

In practice, implementation of INDO translates to construction of a modified CNDO/2 Fock matrix for elements with orbitals from the same atom. In the original paper detailing implementation, which served as our reference, these integrals are derived from 3 new atom-specific empirical parameters. Two of these, G^0 and F^2 , are entirely new. The third, F^0 , is equal the CNDO/2 γ matrix element γ_{AA} . All three were added as new constants that could be accessed via the `get_atom_constant` method of our Atom class. Additionally, we implemented the `get_integral` method to our system class to compute the one-center electron integrals using these constants.

With these enhancements in place, we implemented an alternative calculation of the block-diagonal intra-atom elements of the Fock matrix using the one-center electron integrals discussed

above, as outlined in the INDO method. As a secondary modification, the electronic energy calculation was also updated to account for the inclusion of the new electron integrals in the Fock matrix.

3.3. Semi-Empirical Calculation of ^{13}C Chemical Shifts in Paraffins with Structural Parameters

As discussed previously, Takaishi proposed a highly empirical, but computationally simple, model for calculating magnetic shielding tensors of paraffins, that is, non-cyclic hydrocarbons. As is common for calculations of σ , Takaishi divides this into computations of diamagnetic (σ_d) and paramagnetic (σ_p) components, neglecting other contributions such as those from ring currents. Both diamagnetic and paramagnetic components avoid perturbative methods, instead being computed directly from the ground state configuration of the molecule.

Takaishi's diamagnetic term is purely a quadratic function of the electron density, q , around the atom in question and is given as:

$$\sigma_d = 4.45z^* \cdot q \quad (3)$$

where

$$z^* = 3.25 - 0.35(q - 4) \quad (4)$$

The precise definition of q is unspecified in the paper, and a divergence in our interpretations might be a source of error for our results. In our interpretation, q_A is the expected value of the number of electrons found in any of an atom A 's orbitals, or:

$$q_A = 2 + \sum_{\mu}^{MO_{occupied}} \sum_i^A \langle i|\mu\rangle \langle \mu|i\rangle \quad (5)$$

With the addition of 2 accounting for the core orbitals of carbon that are not included in CNDO/2 or INDO calculations.

The paramagnetic term is given by:

$$\sigma_p^A = -\frac{e^2\hbar^2}{2m^2c^2\Delta E} \left\langle \frac{1}{r^3} \right\rangle_{2p} \sum_B Q_{AB} \quad (6)$$

Notably, this term is dependent on a new parameter, ΔE , the electronic excitation energy of the atom. Takaishi introduces the following equation to calculate ΔE :

$$\Delta E = K \times \alpha^a \times \beta^b \times \gamma^c \times S \quad (7)$$

$$S = [1^\circ(4^\circ)]^d \times [2^\circ(3^\circ)]^e \times [2^\circ(4^\circ)]^f \times [3^\circ(2^\circ)]^g \times [3^\circ(3^\circ)]^h \times [4^\circ(2^\circ)]^i \quad (8)$$

Where a, b, c are the number of α, β , and γ carbons neighboring the primary carbon. Additionally, further local structural information is encoded via S , which contains constants raised to the power of counts of the configuration of neighbors of neighbors of the primary carbon. K is then, in theory, the excitation energy of an electron from the carbon atom of methane. In practice, Takaishi fitted this and the other constants in equation 7 empirically, arriving at a value of $K = 11.30\text{eV}$. We used these values for our implementation.

The equations and constants above fully describe the process used to generate predicted magnetic shielding tensors with our CNDO/2 model. In addition to the above, we added the additional step of minimizing least squares loss via linear regression for our INDO model, as it was not what Takaishi's model had been empirically fitted for. Lastly, we would note that while Takaishi reports all chemical shifts relative to those of benzene, they do not specify in their paper what value they were using for the magnetic shielding tensor of benzene. Therefore, we calibrate our results to that of Takaishi's by adding a constant term to match our methane chemical shift to that of Takaishi.

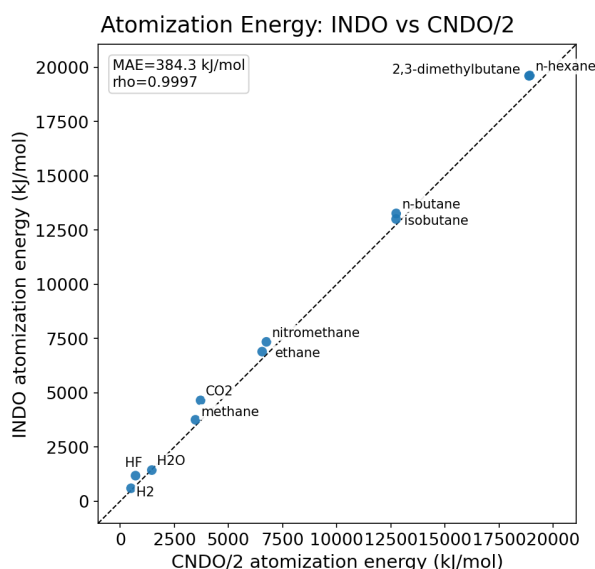


Figure 1. Parity Plots of Atomization Energies for CNDO/2 and INDO methods

4. RESULTS

4.1. Validation of DIIS Implementation

Before applying our core algorithmic and model enhancements to NMR calculations, we first sought to validate their correctness in isolation. Thus, we first performed tests to confirm that our implementation of the DIIS algorithm was accurate and outperformed fixed-point iteration both with respect to numerical stability and number iterations required to converge. In order to assess the algorithmic effectiveness of DIIS, we ran the SCF cycle for each of the molecules we had data for, using both DIIS and fixed-point iteration. For each method, convergence or failure was recorded. In addition, for those systems that did converge, the number of iterations required was tabulated. Full results can be found in 8.

Significantly, out of the 21 molecules we tested, DIIS converged for all them. By contrast, fixed-point iteration failed to converge for nearly half (10) of them. Additionally, in cases where both converged, DIIS consistently required fewer iterations to achieve convergence, although this does come with the caveat that DIIS is also computationally more expensive per iteration. Therefore, all following results were attained using our DIIS implementation.

4.2. Validation of INDO Implementation

Second, we performed validations on our implementation of INDO. Namely, we used it to calculate the atomization energies of the molecules in our dataset. While both CNDO/2 and INDO are known to systematically overestimate atomization energies[4][3], the correlation between true atomization energy and predicted energy should still be high[4][3]. Indeed, this is what we observed, with our CNDO/2 model's predictions having a ρ of 0.9985, and INDO very marginally underperformed with a ρ of 0.9974. This tight correlation is highly suggestive of a functioning implementation of INDO. Parity plots of results can be found in Fig. 1.

4.3. ^{13}C Paraffin Spectra

We evaluated our implementation of Takaishi's method against the reference material provided in table (1) of the paper[5]. Predicted spectra were computed using both the CNDO/2 model, as was used in the reference, as well as the INDO model. Parity plots of the results can be found in Fig. 3. In the case of INDO, for which the parameters were not directly fitted to, we took the additional step of applying a linear regression best fit. Aside from Takaishi's own predictions, we also compared our results with the experimental values used for reference within the paper.

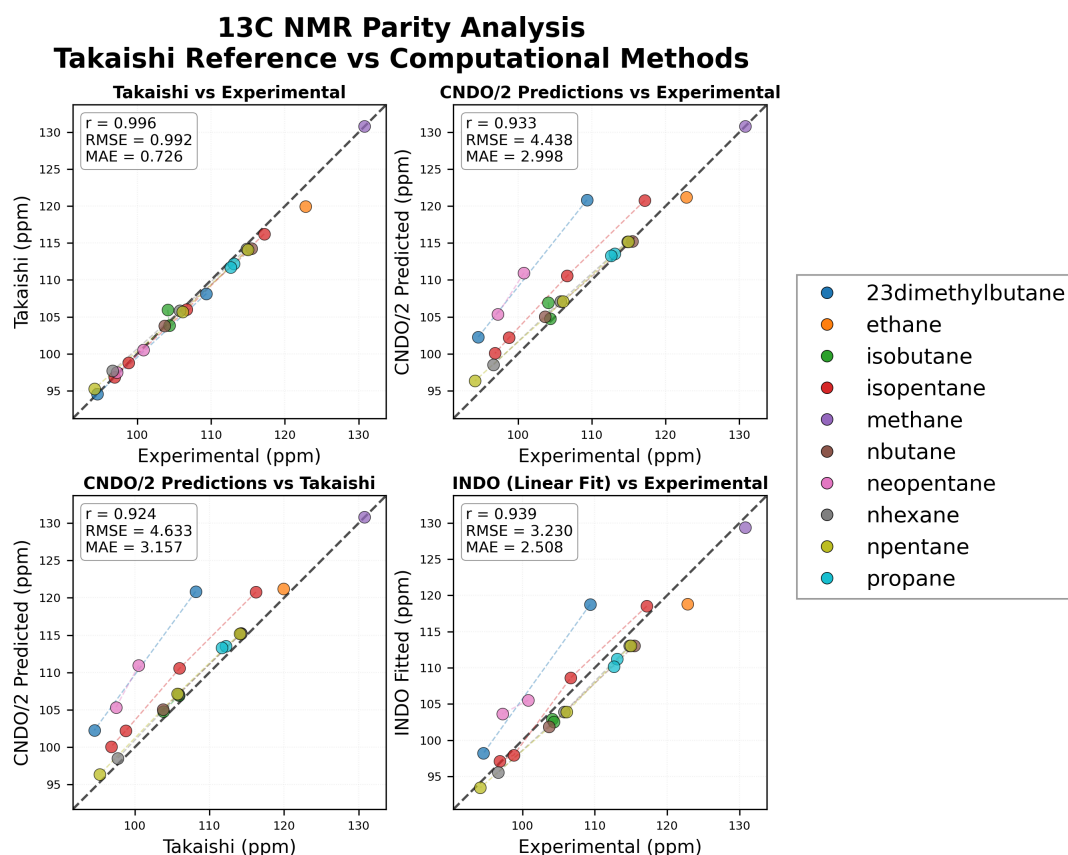


Figure 2. Parity Plots of ^{13}C NRM spectra

Overall, our results reasonably align with those of the reference material. For CNDO/2, we achieve a MAE of 2.998ppm from experimental results (3.157ppm from Takaishi) and a ρ of 0.933 (0.924). For the regression fitted INDO model, we unsurprisingly achieve a superior MAE of 2.508. However, what is more revealing is the regression invariant ρ , which was 0.939, slightly outperforming the CNDO model despite the mismatch in fitted parameters.

A particularly salient example of INDO’s performance despite not being the model fitted to the data is found in the spectra for isobutane. Both our CNDO/2 model and Takaishi’s significantly overestimate the distance between isobutane’s two peaks, however, the INDO model accurately placed them close to one another. In general, the plots suggest that INDO performs better than CNDO/2 at accurately predicting the spacing between consecutive peaks of NMR spectra, if not the relative distance between peaks from different spectra.

4.4. ^1H Spectra

We next turned to the prediction of ^1H NMR spectra using our modified implementation of Popel 1962[1]. For this, we again largely restricted our analysis to linear and branching hydrocarbons, both for the sake of comparison with our ^{13}C results as well as to restrict the diversity of our molecules to allow for a more effective linear regression to be performed. We find relatively strong correlations between predicted spectra and experimental results. To wit, we calculate a value for ρ of 0.867 using our CNDO/2 model, and 0.891 with our INDO model. Again, we see INDO performs slightly better, consistent with its superior modeling of intra-atomic orbital interactions. INDO would likely perform better relative to CNDO/2 if this model were applied to ^{13}C NMR, as it more accurately captures the interactions between same-atom AOs.

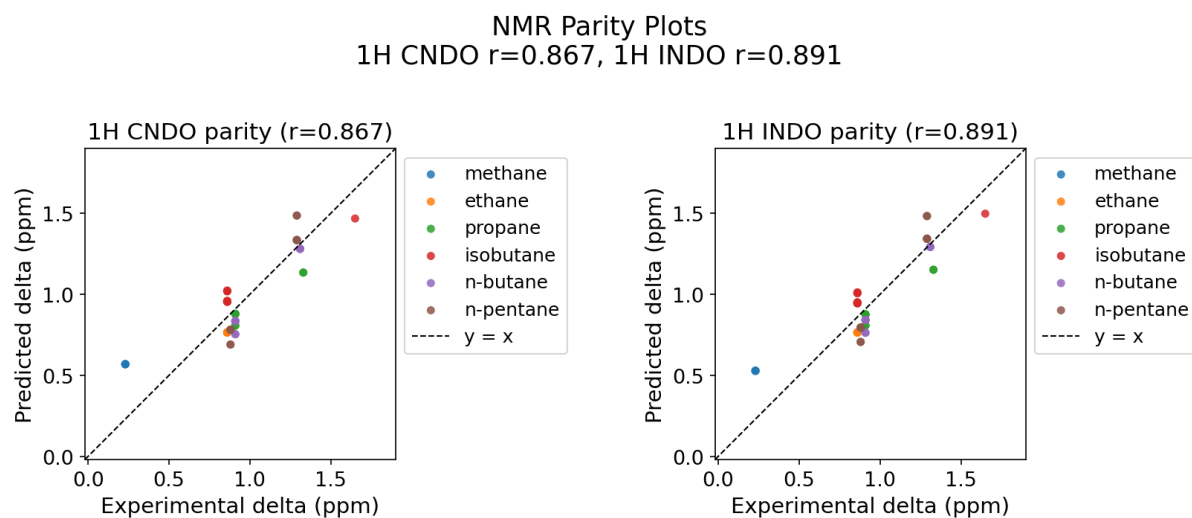


Figure 3. Parity Plots of ^1H NRM spectra

5. CONCLUSION

In this report, we worked to generate accurate NMR spectra for both ^1H and ^{13}C nuclei. We successfully reproduced Takaishi’s method for computing ^{13}C spectra of linear and branching hydrocarbons up to a ρ of about 0.93. Similarly, we achieve concordance with empirical measurements of ^1H with a ρ of around 0.83. It is also clear from our results that fixed-point iteration is insufficient for reliably converging HF SCF cycles of non-trivial molecules. All methods were implemented in a computationally efficient manner, allowing for near-instantaneous evaluation on consumer grade electronics.

6. NEXT STEPS

In the future, we would like to extend our ^1H methodology to ^{13}C NMR. This will allow for direct benchmarking between a highly empirical (Takaishi 1974) and less empirical (Popel 1962) approach to NMR spectra prediction. Additionally, it will allow us to test the hypothesis that the INDO model will perform better than CNDO/2 on second row atoms that have substantial inter-atomic AO interactions.

Additionally, we would like to further build upon on implementation of INDO by expanding it to an implementation of MINDO. MINDO is likely to be substantially more accurate than either INDO or CNDO/2. Moreover, its parameters could be tuned to maximize its accuracy for NMR calculations specifically.

7. CONTRIBUTIONS

Colman provided the foundation for the codebase. He also implemented the DIIS algorithm and the INDO model. Additionally, implemented the diamagnetic term of Takaishi’s model and the paramagnetic term aside from the calculation ΔE . Lastly, he generated the final versions of the plots, and wrote the report.

Alex implemented the calculation of ΔE for the implementation of Takaishi’s model. Additionally, he implemented the σ_d term from Popel 1963, and introduced a complex-valued version of the CNDOSystem class to handle perturbations to compute the σ_p term. Likewise, he implement a variant of DIIS to handle complex-valued fock/density matrices. He also created numerous scripts to help visual the output data.

8. APPENDIX

8.1. Code Reference

The code used in this report may be found at <https://github.com/LorentzFactor/chem-279-final-project>

8.2. DIIS Iteration Results

Table 1. Comparison of Fixed-point and DIIS Convergence Status and Iterations

Molecule	Fixed-point convergence	Fixed-point iterations	DIIS convergence	DIIS iterations
2,3-dimethylbutane	failed	–	converged	8
CO ₂	converged	18	converged	9
H ₂	converged	2	converged	1
H ₂ O	converged	20	converged	12
HF	converged	16	converged	13
HO	converged	17	converged	13
TMS	converged	16	converged	8
benzene	converged	13	converged	7
butadiene	failed	–	converged	17
caffeine	failed	–	converged	48
ethane	converged	12	converged	7
isobutane	converged	16	converged	8
isopentane	failed	–	converged	9
methane	converged	13	converged	4
n_butane	failed	–	converged	8
n_hexane	failed	–	converged	8
n_pentane	failed	–	converged	8
neopentane	converged	21	converged	8
nitroethane	failed	–	converged	26
nitromethane	failed	–	converged	13
propane	converged	14	converged	8

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